

MT8002 Elements of Set Theory

Appendix: Logical Notation, Mathematical Notation and Acronyms

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Preliminaries

Textbook

Derek Goldrei (1996). Classic Set Theory. A Guided Independent Study.

Logical Notation

Logical constants

$\wedge$	(and)	conjunction
$\vee$	(or)	inclusive disjunction [†]
$\rightarrow$	(if __, then __)	conditional, (material) implication
$\neg$	(not)	negation
$\leftrightarrow$	(if and only if)	bi-conditional, (material) equivalence
$\forall x$	(for every x)	universal quantifier
$\exists x$	(there exists a x)	existential quantifier
$\exists! x$	(there exists one and only one x)	unique existential quantifier
$\doteq$	(equal)	equality, identity [‡]

[†]One or the other or both.

[‡]The textbook uses the symbol '='.

Logical Notation

Non-logical symbols

$\in$ (belong) membership relation

Meta-logical symbols

$:=$ (is defined as) left side is being defined by the right side

$=:$ (is defined as) right side is being defined by the left side

Mathematical Notation and Acronyms

Numerical sets

$\mathbb{N}$: Set of natural numbers (including the zero)

$\mathbb{Z}$: Set of all integers (positive, negative and zero)

$\mathbb{R}$: Set of real numbers

Acronyms

FOL (First-Order Logic)

FOLEQ (First-Order Logic with Equality)

ZFC (Zermelo–Fraenkel Set Theory with Choice)

References

Derek Goldrei (1996). Classic Set Theory. A Guided Independent Study. Chapman & Hall (cit. on p. 2).